

Let's build a Time Machine!

Migration from CRUD to ES

01.04.2025

The core of ES - events

- The single source of truth for the system's state
- Good events:  *OrderPlaced, OrderShipped, OrderDelivered*
- Bad events:  *OrderCreated, OrderUpdated, OrderStatusUpdated*
- Explicit, fluent, expressive, and self-contained
- Not CRUD-based but rather business-oriented!

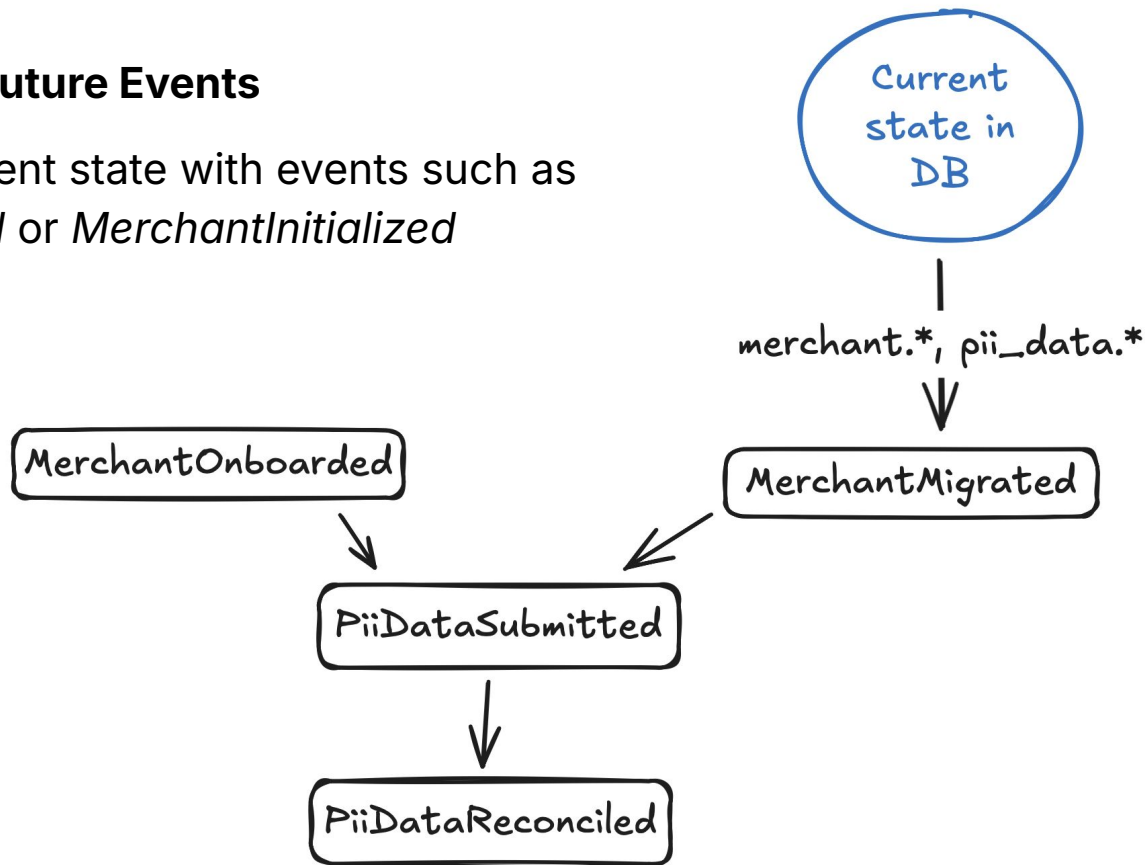


Migration prerequisites:

1. Decide which parts can benefit from ES and which should stay as CRUD
2. Identify ES **entities**, **aggregates** and **boundaries**
 - ⚠ Aggregate is a cluster of entities with a shared identity and consistency boundary
3. Define migration strategy for initial state

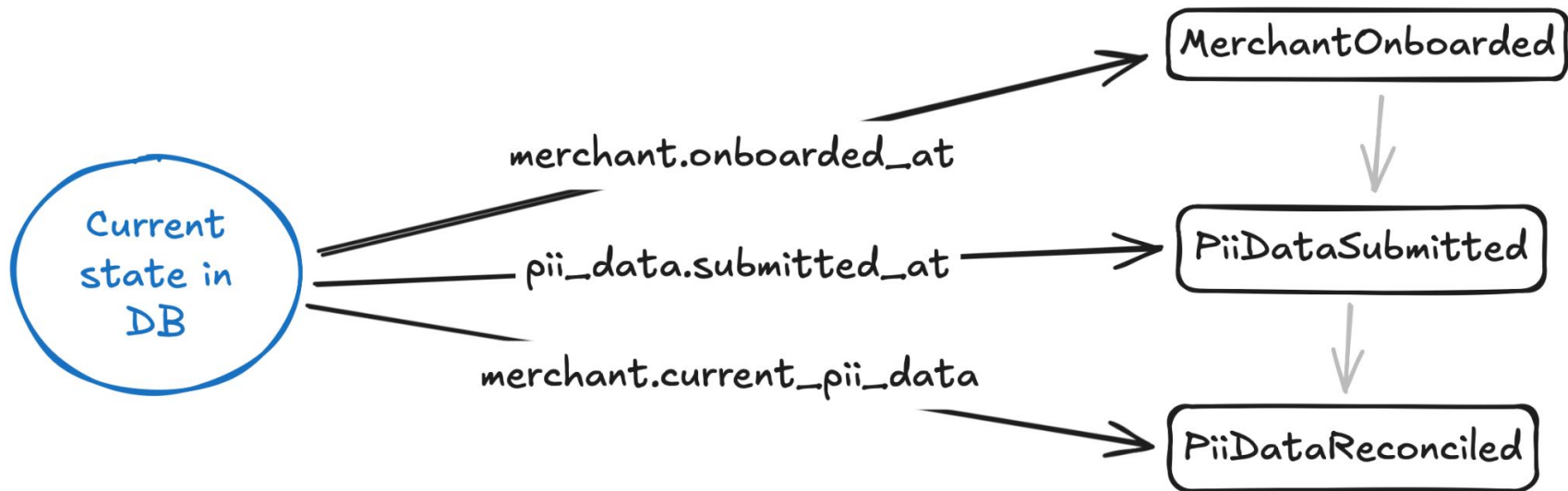
→ Option 1: Snapshot + Future Events

- Introduce the current state with events such as *MerchantMigrated* or *MerchantInitialized*



→ Option 2: Event Capture

- Pay attention to indicators like timestamps or statuses
- Reverse-engineer past events from the current state



Gradual migration strategy:

1. **Start with write model**, take one aggregate at a time and refactor it to use ES
 - Until ready, hide it behind a Feature Flag
 - Before rolling out, initialize streams with migration events
2. For some time, **write to both CRUD and ES** systems (dual-write)
 - Test and validate ES against CRUD
3. **Rebuild projections** (read model) by replaying captured events

It's not all 🦄 and 🌈

- It's impossible to get everything right all the time
- Introduce metrics, and optimize performance as needed
 - Reach for snapshots only as a last option
- Gather feedback from users and developers

